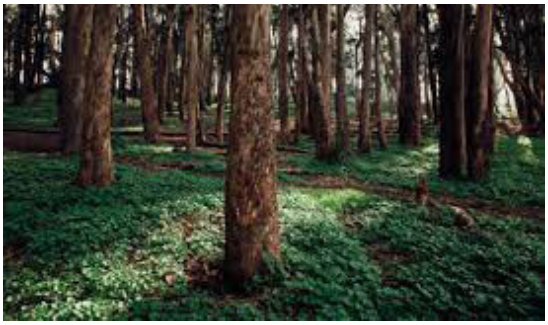
4.2 A Powerful Tool for Overcoming the Limitations of Centralization

Decentralized systems, on the other hand, can be thought of as a forest. Each tree represents an independent participant, and the forest represents the overall system. In this analogy, the forest is more resilient than an individual tree because it does not rely on a single point of failure. If one tree is damaged or diseased, the rest of the forest can continue to thrive. The trees in the forest share the ground, the nutrients, the sun, and the rain.

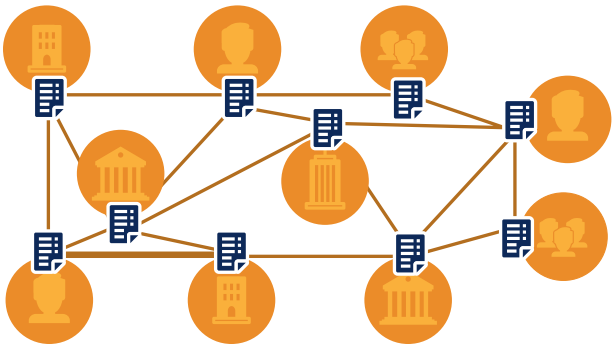


Decentralized systems, like communities, networks, and forests, function best when there is a diverse group of individuals working together, rather than a single central authority dictating all the rules.

A **network** is a group of **nodes** that are connected to each other in some way. This connection allows the devices to exchange information and communicate with each other.

A **node** is a *computer connected to a network* that can *share and/or receive information* and communicate with the other nodes.

Advantages of a decentralized system:

- It is more resilient and reliable because there is ***no single point of failure***. If one part of the system goes down, the rest can continue to operate.
- With the right encryption, decentralization is more ***secure*** because there is no central point of control that can be targeted by hackers.

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